

Solutions to Suggested Problems

- 6.21 $n = 6$ $\mu = 20$ ambulance service claim of mean service time
 $\bar{x} = 2.3$
 $s = 6.39$

$$t = \frac{23 - 20}{6.39/\sqrt{6}} = 1.15$$

$$df = 5$$

$$t < t_{0.1} = 1.476$$

$\therefore$ Claim is reasonable.

- 6.22 $n = 10$ $\mu = 0.5000$ value determines whether process is in control
 $\bar{x} = 0.5060$
 $s = 0.0040$

$$t = \frac{0.5060 - 0.5000}{0.0040/\sqrt{10}} = 4.743$$

$$df = 5$$

$$t > t_{0.001} = 3.250$$

∴ Process is out of control

- 6.24 $n = 10$
 $\sigma^2 = 42.5$

Find $P(3.14 < s < 8.94)$. Let $\chi^2 = \frac{(n-1)s^2}{\sigma^2}$.

$$\frac{9(3.14)^2}{42.5} = 2.088 \qquad \frac{9(8.94)^2}{42.5} = 16.92$$

Find $P(2.088 < \chi^2 < 16.92) = 1 - 0.05 - 0.01 = 0.94$

7.4 $n = 50$
 $\bar{x} = 11,795$
 $s = 14,054$
 $\alpha = 0.05$

maximum error (E) estimate:

$$E = |\bar{x} - \mu| < z_{\frac{0.05}{2}} \frac{14,054}{\sqrt{50}} = 1.960 \frac{14,054}{\sqrt{50}} = 3,896 \text{ with 95\% confidence}$$

7.5 Confidence interval: $(\bar{x} - E, \bar{x} + E) = (11,795 - 3,896, 11,795 + 3,896)$ with 95% confidence.

7.6 $n = 40$
 $\bar{x} = 12.73$
 $s = 2.06$

a) $\alpha = 0.01$

maximum error (E) estimate:

$$E = |\bar{x} - \mu| < z_{\frac{0.01}{2}} \frac{2.06}{\sqrt{40}} = 2.576 \frac{2.06}{\sqrt{40}} = 0.84 \text{ with 99\% confidence}$$

b) $\alpha = 0.02$

$$E_2 = |\bar{x} - \mu| < z_{\frac{0.02}{2}} \frac{2.06}{\sqrt{40}} = 2.326 \frac{2.06}{\sqrt{40}} = 0.76 \text{ with 98\% confidence}$$

Confidence interval: $(\bar{x} - E_2, \bar{x} + E_2) = (12.73 - 0.76, 12.73 + 0.76)$ with 98% confidence.

7.7 $E = 0.5 \text{ min.}$

$$E = |\bar{x} - \mu| < z_{\frac{\alpha}{2}} \frac{2.06}{\sqrt{40}} = 0.5 \qquad z_{\frac{\alpha}{2}} = \frac{0.5 \sqrt{40}}{2.06} = 1.535$$

$$\alpha/2 = 1 - 0.9376 = 0.0624$$

$$\alpha = 0.1248$$

Confidence level = $1 - 0.1248 = 0.8752$ that maximum error is 0.5 min.

7.9 $n = 80$
 $\bar{x} = 472.36$
 $s = 62.35$

$$E = |\bar{x} - \mu| < z_{\frac{\alpha}{2}} \frac{62.35}{\sqrt{80}} = 10$$

$$z_{\frac{\alpha}{2}} = \frac{10 \sqrt{80}}{62.35} = 1.435$$

$$\alpha/2 = 1 - 0.9243 = 0.0757$$

$$\alpha = 0.1514$$

Confidence level = $1 - 0.1514 = 0.8486$ that maximum error is 10.

7.12 $\sigma = 60$
 $1 - \alpha = 0.9$
 $E = 10$

$$E = |\bar{x} - \mu| < z_{\frac{0.1}{2}} \frac{60}{\sqrt{n}} \leq 10$$

$$1.645 \frac{60}{\sqrt{n}} \leq 10$$

$$\sqrt{n} \geq \frac{1.645 (60)}{10}$$

$$n \geq 97.4$$

7.24 $n = 8$

$$E_1 = |\bar{x} - \mu| < t_{\frac{0.05}{2}} \frac{0.54}{\sqrt{8}} = 2.365 \frac{0.54}{\sqrt{8}} = 0.45 \text{ with 98\% confidence}$$

$\bar{x} = 2.1$
 $s = 0.54$

Confidence interval $(\bar{x} - E_1, \bar{x} + E_1) = (2.1 - 0.45, 2.1 + 0.45)$ with 95% confidence.

$1 - \alpha = 0.95$
 $df = 7$

- 7.27 a) Erroneously rejecting the null hypothesis that the dam was safe would cause unneeded renovation and/or repairs.
- b) Erroneously accepting the null hypothesis that the dam was safe would cause loss of property and/or life.

$$7.30 \quad \begin{array}{ll} \mu = 3.0000 & H_0 : \mu = 3.0000 \\ \sigma = 0.0250 & H_1 : \mu \neq 3.0000 \end{array}$$

$$a) \quad n = 30$$

H_0 will be rejected if $|\bar{x} - \mu| > 0.0040$

$$|z| = \frac{|\bar{x} - \mu|}{\sigma/\sqrt{n}} = \frac{0.0040}{0.0250/\sqrt{30}} = 1.012$$

$$P(|z| > 1.012) = 2 P(z > 1.012) = 2(1 - 0.8670) = 0.2660$$

$$b) \quad n = 30$$

Find $P(x < 2.9960) + P(x > 3.0040)$ if $\mu = 3.0050$ prevails

$$z_a = \frac{2.9960 - 3.0050}{0.0250/\sqrt{30}} = -1.972$$

$$z_b = \frac{3.0040 - 3.0050}{0.0250/\sqrt{30}} = -0.219$$

$$\text{Find } P(z < z_a) + P(z > z_b) = 0.02433 + 1 - 0.4125 = 0.6118$$

$$7.32 \quad \begin{array}{l} \mu = 100 \\ \sigma = 12 \\ n = 40 \\ \alpha = 0.01 \end{array}$$

$$\text{Find } \bar{x}_0 \text{ so that } P(x > \bar{x}_0) = 0.01. \quad \text{Let } z_a = \frac{\bar{x}_0 - \mu}{\sigma/\sqrt{n}} = \frac{\bar{x}_0 - 100}{12/\sqrt{40}}.$$

$P(z > z_a) = 0.01$ is equivalent to $z_a > z_{0.01} = 2.326$. Hence

$$\bar{x}_0 > 100 + 2.326 \frac{12}{\sqrt{40}} = 100 + 4.41.$$

7.39 n = 45 μ = 73.2
 $\bar{x} = 76.7$ σ = 8.6

$\alpha = 0.01$

 $H_0: \mu = 7.32$
 $H_1: \mu > 73.2$

 $\alpha = 0.01$

✓ $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ Critical value: $z_{\alpha} = 2.326$

✓ $\frac{76.7 - 73.2}{8.6/\sqrt{45}} = 2.730$

X Reject H_0 . The data provides sufficient evidence that mean aptitude score is more than 73.2.

7.40 n = 35 μ = 1.3
 $\bar{x} = 1.4707$
 s = 0.5235

$\alpha = 0.05$

 $H_0: \mu = 1.3$
 $H_1: \mu > 1.3$

 $\alpha = 0.05$

✓ $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ Critical value: $z_{\alpha} = 1.645$

✓ $\frac{1.4707 - 1.3}{0.5325/\sqrt{35}} = 1.929$

X Reject H_0 . The data provides sufficient evidence that mean cost is more than 1.3 (thousands).

7.41 $n = 64$ $\mu = 1,000$
 $\bar{x} = 1,038$
 $s = 146$

$\alpha = 0.05$

 $H_0: \mu = 1000$
 $H_1: \mu > 1000$

 $\alpha = 0.05$

✓ $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ Critical value: $z_\alpha = 1.645$

✓ $\frac{1,038 - 1,000}{146/\sqrt{64}} = 2.08$

✗ Reject H_0 . The data provides sufficient evidence that mean number of acceptable pieces is more than 1,000.

7.43 $n = 60$ $\mu = 32.6$
 $\bar{x} = 33.8$
 $s = 6.1$

$\alpha = 0.05$

 $H_0: \mu = 32.6$
 $H_1: \mu > 32.6$

 $\alpha = 0.05$

✓ $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ Critical value: $z_\alpha = 1.645$

✓ $\frac{33.8 - 32.6}{6.1/\sqrt{60}} = 1.52$

✗ Fail to reject H_0 . The data does not provide sufficient evidence that mean travel time is more than 32.6 minutes.

7.44 n = 6 μ = 58,000
 $\bar{x} = 58,392$
 s = 648

$\alpha = 0.05$

 $H_0: \mu = 58,000$
 $H_1: \mu \neq 58,000$

 $\alpha = 0.05$

✓ $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$ Critical value: $t_{\alpha/2} = 2.571$
df = 5

$$\checkmark \quad \frac{58,392 - 58,000}{648/\sqrt{5}} = 1.481$$

X Fail to reject H_0 . The data does not provide sufficient evidence that mean compressive strength is not than 58,000 psi.

7.48 n = 5 μ = 14.0
 $\bar{x} = 14.4$
 s = 0.158

$\alpha = 0.05$

 $H_0: \mu = 14.0$
 $H_1: \mu > 14.0$

 $\alpha = 0.05$

✓ $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$ Critical value: $t_{\alpha} = 2.132$
df = 4

$$\checkmark \quad \frac{14.4 - 14.0}{0.158/\sqrt{4}} = 5.660$$

X Reject H_0 . The data provides sufficient evidence that mean tar content is more than 14.0.

$$7.49 \quad n = 5 \quad \mu = 14.0$$

$$\bar{x} = 14.7$$

$$s = 0.742$$

$$\alpha = 0.05$$

$$\begin{aligned} H_0 : & \mu = 14.0 \\ H_1 : & \mu > 14.0 \end{aligned}$$

$$\alpha = 0.05$$

$$\checkmark \quad t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \quad \text{Critical value: } t_{\alpha} = 2.132$$

$$df = 4$$

$$\checkmark \quad \frac{14.7 - 14.0}{0.742/\sqrt{4}} = 2.109$$

✗ Fail to reject H_0 . The data does not provide sufficient evidence that mean tar content is more than 14.0.

The increase in the standard deviation allows for more variability in the population data.

$$7.63 \quad \begin{aligned} \mu_1 &= 0.249 & \mu_2 &= 0.255 \\ \sigma_1 &= 0.003 & \sigma_2 &= 0.002 \end{aligned}$$

$$\mu_{x_1 - x_2} = \mu_1 - \mu_2 = -0.006$$

$$\sigma_{x_1 - x_2} = \sqrt{\sigma_1^2 + \sigma_2^2} = 0.003605$$

$$P(x_1 - x_2 < 0) = P\left(z < \frac{0 - (-0.006)}{0.003605}\right) = P(z < 1.664) = 0.9519$$

$$\begin{array}{ll}
 7.64 & n_1 = 71 \\
 & \bar{x}_1 = 83.2 \\
 & s_1 = 19.3 \\
 & n_2 = 75 \\
 & \bar{x}_2 = 90.8 \\
 & s_2 = 21.4
 \end{array}$$

$$\alpha = 0.05$$

(a)



$$\begin{array}{ll}
 H_0 : & \mu_1 = \mu_2 \\
 H_1 : & \mu_1 \neq \mu_2
 \end{array}$$



$$\alpha = 0.05$$



$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (0)}{\sigma_{\bar{x}_1 - \bar{x}_2}}$$

$$\text{Critical value: } -z_{\alpha/2} = -1.960$$

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{19.3^2}{71} + \frac{21.4^2}{75}} = 3.369$$



$$\frac{(83.2 - 90.8) - 0}{3.369} = -2.2558$$

✗ Reject H_0 . The data provides sufficient evidence that the two population means are different.

(b)

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2} = \sqrt{19.3^2 + 21.4^2} = 28.81$$

$$d = \frac{|\mu_1 - \mu_2|}{\sigma} = \frac{|0 - (-12)|}{28.81} = 0.416$$

From Table 8 (c) the probability of failing to reject the null hypothesis when $\mu_1 - \mu_2 = -12$ prevails is 0.

$$\begin{array}{ll}
 7.65 & n_1 = 60 \\
 & \bar{x}_1 = 292.5 \\
 & s_1 = 15.6
 \end{array}
 \qquad
 \begin{array}{ll}
 & n_2 = 60 \\
 & \bar{x}_2 = 266.1 \\
 & s_2 = 18.2
 \end{array}$$

$$\alpha = 0.01$$

$$\begin{array}{ll}
 \text{✍} & H_0 : \mu_1 - \mu_2 = 20 \\
 & H_1 : \mu_1 - \mu_2 > 20
 \end{array}$$

$$\text{✍} \quad \alpha = 0.01$$

$$\checkmark \quad z = \frac{(\bar{x}_1 - \bar{x}_2) - (20)}{\sigma_{\bar{x}_1 - \bar{x}_2}} \qquad \text{Critical value: } z_\alpha = 2.326$$

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{15.6^2}{71} + \frac{18.2^2}{75}} = 3.095$$

$$\checkmark \quad \frac{(292.5 - 266.1) - 20}{3.095} = 2.068$$

✗ Fail to reject H_0 . The data does not provide sufficient evidence that the mean weekly salaries for men are \$20 more than the mean weekly salaries for women.

$$\begin{array}{ll}
 7.67 & n_1 = 8 \\
 & \bar{x}_1 = 9.67 \\
 & s_1 = 1.81 \\
 & n_2 = 10 \\
 & \bar{x}_2 = 7.43 \\
 & s_2 = 1.48
 \end{array}$$

$$\sigma_1 = \sigma_2 = \sigma$$

$$\alpha = 0.05$$



$$\begin{array}{l}
 H_0 : \mu_1 - \mu_2 = 1.5 \\
 H_1 : \mu_1 - \mu_2 > 1.5
 \end{array}$$



$$\alpha = 0.05$$



$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (1.5)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{Critical value: } t_{\alpha} = 1.746$$

$$df = 16$$

$$s_p^2 = \frac{(n_1 - 1) s_1^2 + (n_2 - 1) s_2^2}{n_1 + n_2 - 2} = \frac{7(1.81)^2 + 9(1.48)^2}{16} = 2.665$$



$$\frac{(9.67 - 7.43) - 1.5}{1.633 \sqrt{\frac{1}{8} + \frac{1}{10}}} = 0.9556$$



Fail to reject H_0 . The data does not provide sufficient evidence that the mean deniers from the two spinning machines differ by more than 1.5.

$$\begin{array}{ll}
 7.68 & n_1 = 10 \\
 & \bar{x}_1 = 70 \\
 & s_1 = 3.37 \\
 & n_2 = 10 \\
 & \bar{x}_2 = 74 \\
 & s_2 = 5.40
 \end{array}$$

$$\sigma_1 = \sigma_2 = \sigma$$

$$\alpha = 0.05$$



$$\begin{array}{l}
 H_0 : \mu_1 = \mu_2 \\
 H_1 : \mu_1 < \mu_2
 \end{array}$$



$$\alpha = 0.05$$



$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (0)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\text{Critical value: } -t_{\alpha} = -1.734$$

$$df = 18$$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{9(3.37)^2 + 9(5.40)^2}{18} = 20.258$$



$$\frac{(70 - 74) - 0}{4.50 \sqrt{\frac{1}{10} + \frac{1}{10}}} = -1.9876$$



Reject H_0 . The data provides sufficient evidence that the training under Method B is more effective.

$$\begin{array}{ll}
 7.69 & n_1 = 9 \\
 & \bar{x}_1 = 58 \\
 & s_1 = 10.4 \\
 & n_2 = 6 \\
 & \bar{x}_2 = 51.8 \\
 & s_2 = 12.7
 \end{array}$$

$$\sigma_1 = \sigma_2 = \sigma$$

$$\alpha = 0.01$$



$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$



$$\alpha = 0.01$$



$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (1.5)}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\text{Critical value: } t_{\alpha/2} = 3.012$$

$$df = 13$$

$$s_p^2 = \frac{(n_1 - 1) s_1^2 + (n_2 - 1) s_2^2}{n_1 + n_2 - 2} = \frac{8(10.4)^2 + 5(12.7)^2}{13} = 128.594$$



$$\frac{(58 - 51.8) - 0}{11.34 \sqrt{\frac{1}{9} + \frac{1}{6}}} = 1.037$$

✗ Fail to reject H_0 . The data does not provide sufficient evidence that the means from the two populations are different.

7.71 $n = 10$
 $\bar{d} = -0.02$
 $s_d = 0.0287$

$\alpha = 0.05$

 $H_0: \bar{d} = 0$
 $H_1: \bar{d} \neq 0$

 $\alpha = 0.05$

✓ $t = \frac{\bar{d} - 0}{s_d / \sqrt{n}}$ Critical value: $-t_{\alpha/2} = -2.262$
 $df = 9$

✓ $\frac{-0.02 - 0}{0.0287 / \sqrt{10}} = -2.2059$

✗ Fail to reject H_0 . The data does not provide sufficient evidence that the two scales weigh differently.